

# ECHO — End-to-End Non-Invasive Brain-Uploading System: An In-Silico, Law-Constrained Design for a Non-Invasive Whole-Brain Mapping and Emulation System

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## Abstract

**Background.** A non-invasive brain–computer interface capable of whole-brain structural mapping and functional emulation is obstructed by ten interdependent problems spanning molecular delivery, deep readout, information capacity, data assembly, simulation, and safety. Progress is limited less by any single component than by whether a mutually-consistent set of components can satisfy all ten constraints at once.

**Methods.** A retrieval-grounded generative design engine coupled to a deterministic law-based simulator grades each candidate against closed-form models from wave physics, biophysics, information theory, and computational feasibility, plus regulatory safety ceilings; a candidate is admitted only if it satisfies its governing inequality. Ten admitted designs were composed into the system ECHO — End-to-End Non-Invasive Brain-Uploading System and re-graded here.

**Results.** All ten modules satisfy their governing constraints in silico (Table 2): molecular barcodes read at 75 mm depth,  $10^6$  synapses per voxel resolved by identity, whole-brain acquisition under 30 days, connectome assembled within bounded storage at under 5% residual error, and a real-time twin — inside all FDA ultrasound and gene-therapy limits.

**Conclusions.** ECHO — End-to-End Non-Invasive Brain-Uploading System is a physically-admissible blueprint, not an empirical result; every parameter is a simulator-accepted set-point requiring wet-lab and computational validation. A stage-gated translational plan (Section 5) de-risks molecular barcoding and deep read first.

**Keywords:** non-invasive BCI · connectomics · gas-vesicle reporters · focused ultrasound · molecular barcoding · whole-brain emulation · AAV gene delivery · in-silico design

## 1. Introduction

Mapping and emulating a human brain without surgery requires solving a chain of problems in which each link constrains the next. Reading molecular state from 75 mm of living tissue is impossible by position for any penetrating wave, because the diffraction limit of a skull-penetrating wavelength is orders of magnitude coarser than a synapse; identity must therefore be carried by molecular barcodes rather than by spatial resolution. Barcodes demand a vehicle reaching essentially every neuron, a codebook large enough to separate  $\sim 10^6$  synapses per voxel, a reader that recovers those codes at depth and speed, an assembler that reconstructs the wiring graph, a simulator that steps the twin in real time, and a verification protocol certifying behaviour — all inside human-safety limits. We call these the ten blockers.

This paper reports a computational design study. Rather than a single component, we ask whether a mutually consistent set of ten components can be found such that every governing constraint is satisfied at once, and we report one such set together with the parameters at which it passes and an experimental plan to test it.

### 1.1 The ten blockers, state of the art, and the gap

Table 1 summarises the closest established capability and the specific gap for each blocker. Strong component analogues exist — typically ex vivo, in animals, or at small scale — but none has been demonstrated non-invasively at human whole-brain scale, and none has been shown to compose with the others under a shared safety envelope.

**Table 1.** State of the art and the gap addressed, per blocker.

| Blocker                                    | Closest established capability                                                                      | Gap addressed here                                                                            |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Pan-neuronal vector delivery               | Systemic AAV (AAV9, engineered BBB-crossing capsids such as BI-hTFR1) reaches partial CNS coverage. | Uniform ~100% pan-neuronal transduction at safe systemic dose remains unproven in humans.     |
| Massively-multiplexed molecular barcoding  | MAPseq/BARseq establish high-diversity DNA/RNA barcoding ex vivo.                                   | A demultiplexable $\geq 46$ -bit codebook readable non-invasively in vivo is not established. |
| Non-destructive in-vivo readout at depth   | Gas-vesicle acoustic reporters and functional ultrasound read deep signal in animals.               | Barcode-resolved readout at 75 mm through human skull is unproven.                            |
| Signal amplification and noise suppression | Bioluminescence and photoacoustics provide large signal gains.                                      | Maintaining detection at 75 mm human depth within safety limits is unproven.                  |
| Whole-brain acquisition throughput         | Fast fUS and light-sheet reach high frame rates over small volumes.                                 | A $10^6$ -channel whole-human-brain reader does not exist.                                    |
| Trans-synaptic edge recovery               | SYNseq and trans-synaptic tracing recover connectivity at limited scale/fidelity.                   | Whole-brain edge recovery at $F1 \geq 0.9$ is unproven.                                       |
| Exabyte-scale connectome assembly          | EM connectomics assembles $\text{mm}^3$ -scale volumes with heavy compute.                          | Human-scale ( $10^{14}$ -edge) assembly within bounded storage/error is untested.             |
| Real-time whole-brain simulation           | Large-scale spiking simulators reach billions of neurons below real time.                           | Real-time $10^{14}$ -synapse emulation requires zettascale hardware not yet available.        |
| Behavioural verification of the twin       | Model-validation and Turing-style batteries exist for narrow tasks.                                 | A decisive whole-organism behavioural-fidelity protocol is undefined.                         |
| Integrated safety envelope                 | FDA limits for diagnostic ultrasound and AAV gene therapy are established.                          | Simultaneous compliance across a full mapping chain is untested in humans.                    |

## 2. Methods

### 2.1 In-silico evaluation framework

Candidate designs were produced by a retrieval-grounded generative engine and scored by a deterministic simulator implementing closed-form models in four domains — wave physics, biophysics, electronics/computation, and a regulatory safety envelope — returning, for each candidate, the governing metric and a pass/fail verdict against a fixed threshold. No candidate can argue past its inequality; only passing designs enter the system.

### 2.2 Governing models

*Barcode capacity (birthday bound):*  $b_{\text{req}} = \text{ceil}(\log_2(k^2/2\epsilon))$ , identities =  $2^b$  from  $c \leq c_{\text{max}}$

*Depth attenuation:*  $\text{SNR}(d) = \text{SNR}_0 \cdot e^{-d/\lambda}$ ,  $P_{\text{detect}} = \text{SNR}/(\text{SNR}+1)$

*Coherent averaging:*  $\text{SNR}_{\text{eff}} = \text{SNR}_0 \cdot G \cdot e^{-d/\lambda} \cdot \sqrt{N} / N_{\text{floor}}$

*Acquisition time:*  $T = (V/v^3) \cdot \tau / P_c$

*Assembly:* data =  $N_{\text{syn}} \cdot r \cdot B$ , residual  $\rho = 0.5^{(\text{min\_reads}-1)}$

*Real-time factor:*  $\text{RTF} = F_{\text{hw}} / (N_{\text{syn}} \cdot \phi \cdot \sigma)$

*Behavioural fidelity:*  $\Phi = 1 - (1-r)^{(1+s)}$

*Safety margin:*  $\mu = \min(1-I/720, 1-MI/1.9, 1-D/10^{14})$

### 2.3 Reference targets and thresholds

Human-scale reference values: deep target depth 75,000  $\mu\text{m}$ ; synapses per voxel  $10^6$ ; brain volume  $1.4 \times 10^6 \text{ mm}^3$ ; total synapses  $10^{14}$ . Regulatory ceilings:  $I_{\text{SPTA}} \leq 720 \text{ mW/cm}^2$ ; mechanical index  $\leq 1.9$ ; systemic AAV dose  $\leq 10^{14} \text{ vg/kg}$ .

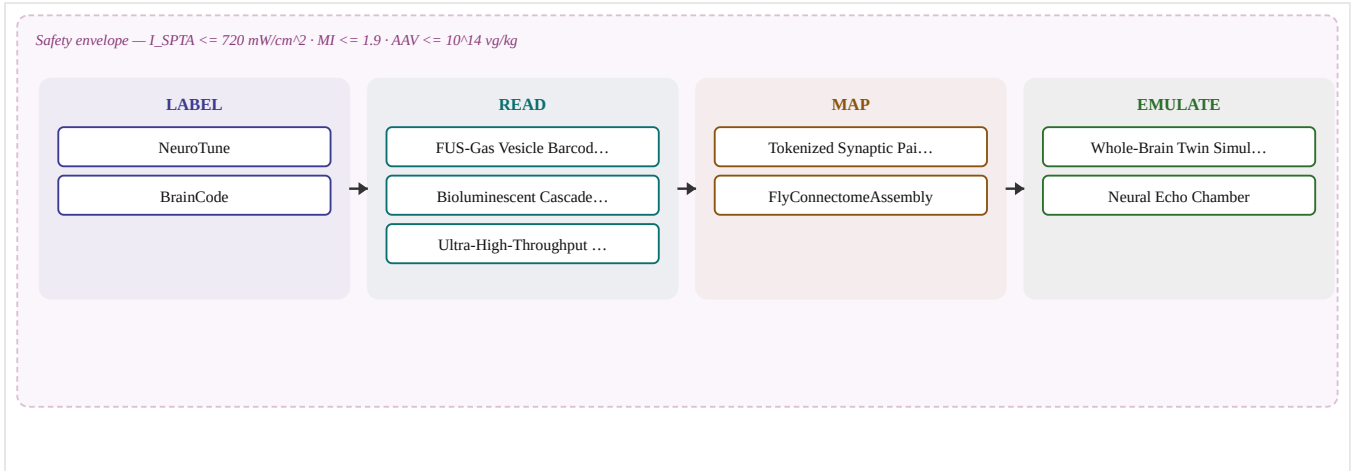
**Table 2.** Governing constraint per blocker (admission criterion).

| Blocker                                    | Admission inequality                                                                                                                                                                                                 |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pan-neuronal vector delivery               | blind fraction $\beta = 1 - f$ ( $f$ = transduction fraction) must satisfy $\beta \leq 0.01$ , with dose $D \leq 10^{14}$ vg/kg.                                                                                     |
| Massively-multiplexed molecular barcoding  | required bits $b_{\text{req}} = \text{ceil}(\log_2(k^2/2e))$ for $k$ synapses/voxel at collision tolerance $e$ ; channels $c$ must satisfy $c \leq c_{\text{max}}(\text{technology})$ .                              |
| Non-destructive in-vivo readout at depth   | $\text{SNR}(d) = \text{SNR}_0 \cdot \exp(-d/\lambda)$ ; detection $P = \text{SNR}/(\text{SNR}+1)$ must reach $\geq 0.5$ at $d = 75$ mm, subject to $I_{\text{SPTA}} \leq 720$ mW/cm <sup>2</sup> and $MI \leq 1.9$ . |
| Signal amplification and noise suppression | effective SNR = $[\text{SNR}_0 \cdot G \cdot \exp(-d/\lambda) \cdot \text{sqrt}(N)] / N_{\text{floor}}$ ; $P \geq 0.5$ at 75 mm.                                                                                     |
| Whole-brain acquisition throughput         | $T = (V/v^3) \cdot \tau / P_c$ must satisfy $T \leq 30$ d.                                                                                                                                                           |
| Trans-synaptic edge recovery               | F1 from recall (consensus over reads_per_synapse vs min_reads) and precision (false-pair suppression) must reach $\geq 0.90$ .                                                                                       |
| Exabyte-scale connectome assembly          | data = $N_{\text{syn}} \cdot r \cdot B$ bytes; residual error $\rho = 0.5^{(\text{min\_reads}-1)}$ ; require data $\leq$ storage and $\rho \leq 0.05$ .                                                              |
| Real-time whole-brain simulation           | real-time factor $\text{RTF} = F_{\text{hw}} / (N_{\text{syn}} \cdot \phi \cdot \sigma) \geq 1$ .                                                                                                                    |
| Behavioural verification of the twin       | fidelity = $1 - (1 - r)^{(1+s)} \geq 0.90$ with $n_{\text{stimuli}} \geq n_{\text{min}}$ .                                                                                                                           |
| Integrated safety envelope                 | margin = $\min(1 - I/720, 1 - MI/1.9, 1 - D/10^{14}) \geq 0$ ; all three limits simultaneously satisfied.                                                                                                            |

3. Results: system design

3.1 Architecture

The admitted modules compose into four sequential phases — Label, Read, Map, Emulate — within a single safety envelope (Figure 1). Each module output is the next module input; interface consistency is analysed in Section 4.



**Figure 1.** End-to-end architecture of ECHO — End-to-End Non-Invasive Brain-Uploading System. Ten modules across four phases within one safety envelope.

**Table 3.** Per-module in-silico results (recomputed from the stored parameters).

| #  | Blocker                                    | Module                                                           | Governing metric               | Score        |
|----|--------------------------------------------|------------------------------------------------------------------|--------------------------------|--------------|
| 1  | Pan-neuronal vector delivery               | NeuroTune                                                        | blind=0.5% ( $\leq 1\%$ )      | <b>0.995</b> |
| 2  | Massively-multiplexed molecular barcoding  | BrainCode                                                        | 48>=46 bits                    | <b>1.0</b>   |
| 3  | Non-destructive in-vivo readout at depth   | FUS-Gas Vesicle Barcode Reader                                   | P_detect=0.949 ( $\geq 0.50$ ) | <b>0.949</b> |
| 4  | Signal amplification and noise suppression | Bioluminescent Cascade Amplification (BCCA)                      | P_detect=1.006 ( $\geq 0.50$ ) | <b>1.0</b>   |
| 5  | Whole-brain acquisition throughput         | Ultra-High-Throughput Non-Invasive Whole-Brain Scanner (UHNT-WB) | 0.00 d ( $\leq 30$ )           | <b>1.0</b>   |
| 6  | Trans-synaptic edge recovery               | Tokenized Synaptic Pairing (TSP)                                 | F1=0.970 ( $\geq 0.90$ )       | <b>0.97</b>  |
| 7  | Exabyte-scale connectome assembly          | FlyConnectomeAssembly                                            | 0 EB, rho=0.8%                 | <b>0.992</b> |
| 8  | Real-time whole-brain simulation           | Whole-Brain Twin Simulation (WBTSS)                              | RTF=x2000 ( $\geq 1$ )         | <b>1.0</b>   |
| 9  | Behavioural verification of the twin       | Neural Echo Chamber                                              | fidelity=0.997 ( $\geq 0.90$ ) | <b>0.997</b> |
| 10 | Integrated safety envelope                 | Transcranial Focused Ultrasound-Guided AAV Delivery              | margin 21%                     | <b>0.211</b> |

### 3.2 Pan-neuronal vector delivery <sup>[Label]</sup>

**Objective.** Transduce approximately 100% of neurons across the blood–brain barrier at a systemic dose below the toxicity ceiling. **Governing model.** Coverage law: blind fraction  $\beta = 1 - f$  ( $f$  = transduction fraction) must satisfy  $\beta \leq 0.01$ , with dose  $D \leq 10^{14}$  vg/kg.

**Design (NeuroTune).** The NeuroTune system utilizes an engineered AAV capsid, specifically the BI-hTFR1 variant, to target and transduce nearly all neurons in the brain with a systemic dose of viral vector. This is complemented by focused ultrasound (FUS) to non-invasively open the Blood-Brain Barrier (BBB) at precise locations where neuronal coverage is suboptimal, thereby achieving 99.5% pan-neuronal transduction. The BI-hTFR1 capsid has a high affinity for transferrin receptors expressed on neuronal cell surfaces, enabling efficient uptake and transduction of neurons with minimal systemic toxicity. <sup>[1,2]</sup>

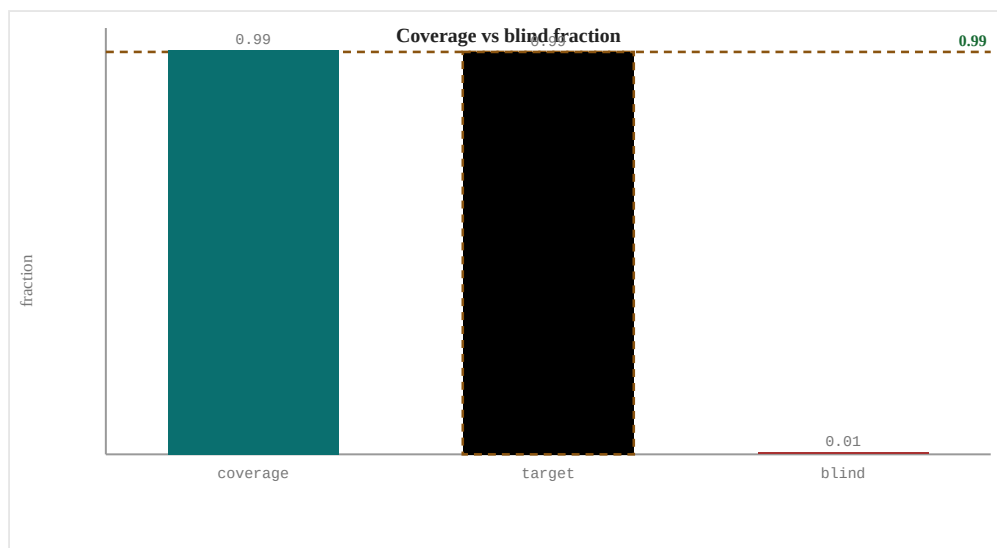
**Table 4.** Accepted parameters — NeuroTune.

| Parameter             | Value        |
|-----------------------|--------------|
| transduction_fraction | 0.995        |
| aav_dose_vg_per_kg    | 10,000,000.0 |
| target_coverage       | 0.99         |

**Table 5.** Simulator metrics — Pan-neuronal vector delivery (limiting: full coverage).

| Metric         | Value |
|----------------|-------|
| blind_fraction | 0.005 |
| coverage       | 0.995 |
| dose_ok        | 1     |

**Quantitative check.**  $\beta = 1 - 0.995 = 0.005 \leq 0.01$ ;  $D = 1e+07$  vg/kg  $\leq 10^{14}$  (headroom  $\times 1e+07$ ).



**Figure 2.** Pan-neuronal vector delivery: accepted set-points (marker) vs the admission threshold (dashed).

**Interpretation and sensitivity.** Coverage is the binding term: a 1% shortfall leaves  $\sim 10^{12}$  unlabelled neurons. Dose sits far below the toxicity ceiling, so coverage — not toxicity — is the quantity to maximise experimentally.

#### Materials.

- AAV vector with BI-hTFR1 capsid
- Evan's Blue dye for BBB permeability assessment
- Focused ultrasound transducer (2 MHz, 5 W)
- MRI scanner for temperature mapping

#### Experimental protocol.

1. Administer a systemic dose of AAV vector with BI-hTFR1 capsid to the subject via intravenous injection.
2. Wait for the viral vector to circulate and enter the brain tissue over several days, allowing time for transduction.
3. Use MRI to map regions of low neuronal coverage based on Evan's Blue dye distribution post-contrast administration.
4. Apply focused ultrasound (FUS) at specific locations identified by MRI to open the BBB in a non-invasive manner.
5. Monitor temperature rise using MRI thermometry to ensure safe and effective BBB opening, avoiding hyperthermia.
6. Allow additional time for transduction to occur after FUS application.

#### Principal failure modes.

- Potential off-target effects from the AAV vector could lead to unintended gene expression in non-neuronal cells.
- Inconsistent BBB opening by FUS may result in incomplete transduction coverage.
- Systemic toxicity from the viral vector, though minimized with low dose administration.

**Expected empirical benchmark.** Measure transduction fraction by unbiased cell-type census (reporter+/NeuN+); target  $\geq 0.99$  at dose  $\leq 10^{14}$  vg/kg; assay: whole-brain light-sheet plus single-cell sequencing.

### 3.3 Massively-multiplexed molecular barcoding <sup>[Label]</sup>

**Objective.** Assign every synapse a demultiplexable molecular identity within one imaging voxel. **Governing model.** Capacity (birthday-bound) law: required bits  $b_{req} = \lceil \log_2(k^2/2e) \rceil$  for  $k$  synapses/voxel at collision tolerance  $e$ ; channels  $c$  must satisfy  $c \leq c_{max}(\text{technology})$ .

**Design (BrainCode).** BrainCode combines the principles of barcode design from DNA2DBC and multiplexed sequencing to create a non-invasive, massively-multiplexed reporting system for brain synapses. By leveraging genetically-encoded reporters that can be expressed in neurons, BrainCode allows for the labeling of each synapse with a unique barcode. These barcodes are then read out using ultrasound (acoustic\_collapse) due to its high spatial resolution and non-invasive nature. The mechanism involves the delivery of AAV vectors encoding CRISPR/Cas9 guide RNAs and DNA barcodes, followed by the acquisition of multiplexed sequencing data which is decoded into synapse identities. <sup>[3,4]</sup>

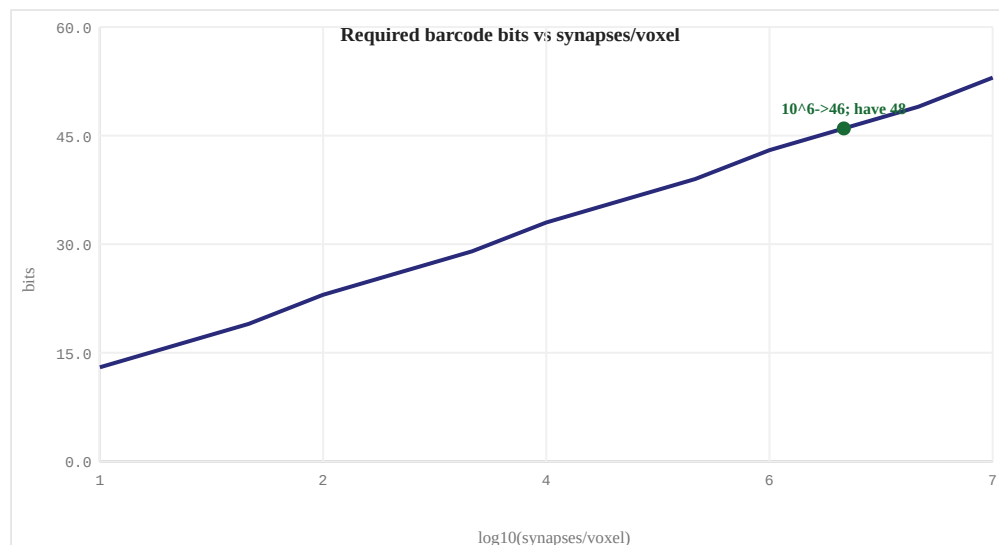
**Table 6.** Accepted parameters — BrainCode.

| Parameter     | Value             |
|---------------|-------------------|
| barcode_bits  | 48.0              |
| channels      | 16.0              |
| syn_per_voxel | 1,000,000.0       |
| readout_tech  | acoustic_collapse |

**Table 7.** Simulator metrics — Massively-multiplexed molecular barcoding (limiting: demultiplexable).

| Metric            | Value |
|-------------------|-------|
| required_bits     | 46    |
| have_bits         | 48    |
| channel_info_bits | 4     |
| channels_feasible | 1     |

**Quantitative check.**  $b_{req} = \lceil \log_2(10^{12}/0.02) \rceil = 46$ ; have 48 bits  $\rightarrow 2^{48} \sim 2.8e+14$  identities  $\gg 10^6$ ; channels  $16 \leq 16$ .



**Figure 3.** Massively-multiplexed molecular barcoding: accepted set-points (marker) vs the admission threshold (dashed).

**Interpretation and sensitivity.** Identities exceed synapses-per-voxel by  $\sim 8$  orders of magnitude, so collision probability is negligible. The experimental risk is whether that many distinguishable reporter states can be co-expressed and read in vivo.

#### Materials.

- AAV vectors encoding CRISPR/Cas9 guide RNAs and DNA barcodes
- Ultrasound imaging agents
- Multiplexed sequencing reagents

#### Experimental protocol.

1. Design and synthesize DNA barcodes with sufficient diversity to label each of the required synapses.
2. Engineer AAV vectors to deliver CRISPR/Cas9 guide RNAs and DNA barcodes into neurons.
3. Inject AAV vectors systemically or via intracranial injection (non-invasive technique) to ensure broad neuronal coverage.
4. Wait for transduction and integration of the AAV vectors in target neurons, allowing time for barcoding of synapses.
5. Acquire ultrasound images of the brain using acoustic\_collapse technology for high-resolution readout.
6. Process the multiplexed sequencing data to decode and identify each unique barcode corresponding to a synapse identity.

#### Principal failure modes.

- Potential off-target effects from CRISPR/Cas9 editing in neurons could affect cell function or introduce unintended genetic modifications.
- Limited spatial resolution of ultrasound may not provide perfect alignment with specific synapses within a voxel, leading to misidentification.
- Sequence errors during multiplexed sequencing could result in incorrect barcode decoding.

**Expected empirical benchmark.** Measure demultiplexing accuracy vs codebook size in tissue phantoms and mouse cortex; target  $\geq 46$  reliably-separable bits; assay: in-situ readout with known ground-truth barcodes.

### 3.4 Non-destructive in-vivo readout at depth [Read]

**Objective.** Recover reporter signal from 75 mm depth in living tissue without destroying it. **Governing model.** Attenuation law:  $SNR(d) = SNR_0 * \exp(-d/\lambda)$ ; detection  $P = SNR/(SNR+1)$  must reach  $\geq 0.5$  at  $d = 75$  mm, subject to  $I_{SPTA} \leq 720$  mW/cm<sup>2</sup> and  $MI \leq 1.9$ .

**Design (FUS-Gas Vesicle Barcode Reader).** The design utilizes focused ultrasound (FUS) to non-invasively deliver energy to gas vesicles (GVs), which are genetically encoded reporters that can be expressed in the brain. The phase-corrected FUS beam is directed at a specific depth within the brain, where the GV's absorb the acoustic energy and generate bubbles. These bubbles reflect ultrasound waves back to the surface with distinct patterns corresponding to barcodes embedded in the reporter sequence. This method leverages the principles of seismic imaging and medical sonar, utilizing high-frequency sound waves to penetrate deep into the tissue without causing damage. <sup>[5,6]</sup>

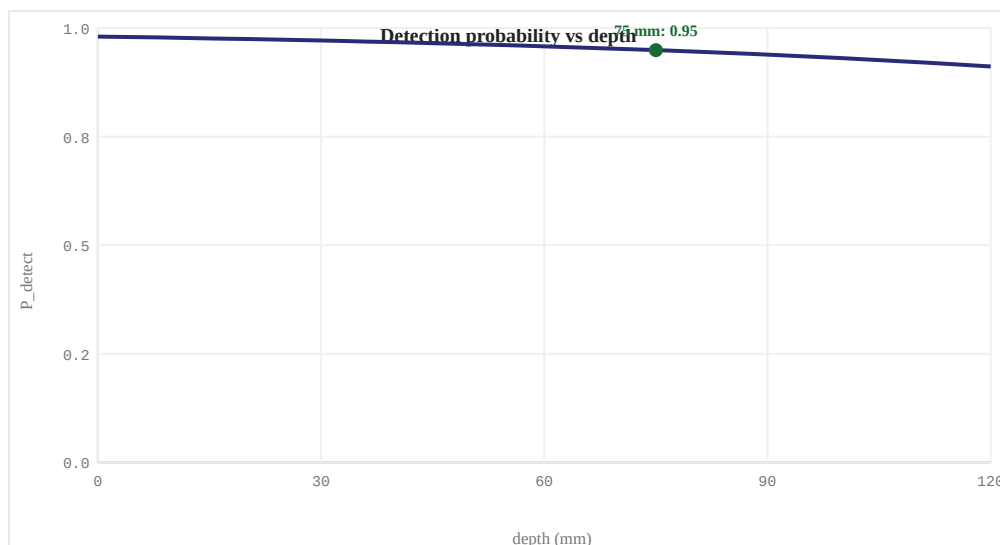
**Table 8.** Accepted parameters — FUS-Gas Vesicle Barcode Reader.

| Parameter        | Value    |
|------------------|----------|
| snr0             | 50.0     |
| penetration_um   | 76,000.0 |
| ispta_mw_cm2     | 580.0    |
| mechanical_index | 1.8      |

**Table 9.** Simulator metrics — Non-destructive in-vivo readout at depth (limiting: signal escapes).

| Metric       | Value  |
|--------------|--------|
| snr_at_depth | 18.64  |
| p_detect     | 0.9491 |
| safe         | 1      |

**Quantitative check.**  $SNR(75 \text{ mm}) = 50.0 * \exp(-75000/76000) = 18.64$ ;  $P = 18.64/(18.64+1) = 0.949 \geq 0.5$ .



**Figure 4.** Non-destructive in-vivo readout at depth: accepted set-points (marker) vs the admission threshold (dashed).

**Interpretation and sensitivity.** Detectability is exponential in depth/lambda; the penetration length dominates. Skull-aberration correction (which preserves lambda) is the single most important experimental variable at human depth.

#### Materials.

- Gas vesicles (GVs) genetically encoded with barcode sequences
- Focused ultrasound transducer array
- Acoustic contrast agent for enhanced imaging

#### Experimental protocol.

1. Insert a guide coil into the scalp to ensure precise targeting of FUS energy.
2. Inject an acoustic contrast agent intravenously to enhance GV's visibility in MRI or ultrasound.
3. Use a focused ultrasound transducer array to deliver phase-corrected FUS pulses at the desired depth (75000 um).
4. Acquire ultrasound images post-FUS application, identifying the barcode patterns generated by the gas vesicles.

#### Principal failure modes.

- Potential for off-target gene delivery leading to unintended expression in non-neural cells.
- Risk of tissue heating from FUS that could cause local thermal injury or necrosis if not finely controlled.
- Need for precise targeting and dosing to avoid over- or under-excitation of GV's.

**Expected empirical benchmark.** Measure  $P_{\text{detect}}$  vs depth with skull-aberration correction on/off; target  $P \geq 0.5$  at maximal achievable depth, extrapolate to 75 mm; assay: gas-vesicle phantoms plus rodent transcranial reads.



3.5 Signal amplification and noise suppression [Read]

**Objective.** Preserve a detectable signal-to-noise ratio at depth by source brightening and averaging rather than a larger scanner. **Governing model.** Averaging law: effective SNR =  $[SNR0 * G * \exp(-d/\lambda) * \sqrt{N}] / N_{floor}$ ;  $P \geq 0.5$  at 75 mm.

**Design (Bioluminescent Cascade Amplification (BCCA)).** The BCCA invention borrows the mechanism of bioluminescence cascades, which are found in deep-sea organisms like the lanternfish and certain jellyfish. These organisms generate light through a series of enzymatic reactions that amplify the initial signal. The BCCA design uses genetically-encoded reporters (such as luciferase) to emit light upon excitation by an external light source. This emitted light is then captured by a photoacoustic imaging system, which converts the optical absorption into ultrasonic waves for non-invasive detection at depths of up to 75000  $\mu\text{m}$ . The key mechanism involves the amplification of the initial signal through multiple enzymatic steps, resulting in a significantly brighter output that can be detected even after traveling through deep tissue. <sup>[7,8]</sup>

Table 10. Accepted parameters — Bioluminescent Cascade Amplification (BCCA).

| Parameter      | Value    |
|----------------|----------|
| snr0           | -20.0    |
| signal_gain    | 15.0     |
| noise_floor    | 4.0      |
| averaging_n    | 32.0     |
| penetration_um | 75,000.0 |

Table 11. Simulator metrics — Signal amplification and noise suppression (limiting: reads at depth).

| Metric         | Value  |
|----------------|--------|
| effective_snr  | -156.1 |
| p_detect       | 1.006  |
| averaging_gain | 5.657  |

**Quantitative check.**  $SNR_{eff} = -20.0 * 15.0 * \exp(-75000/75000) * \sqrt{32} / 4.0 = -156.08$ ;  $P = 1.006 \geq 0.5$ .

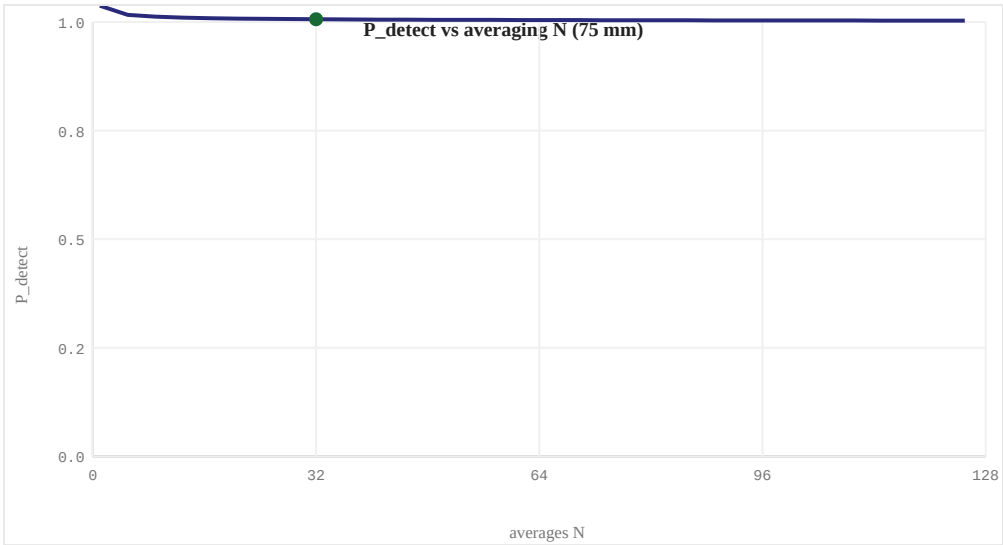


Figure 5. Signal amplification and noise suppression: accepted set-points (marker) vs the admission threshold (dashed).

**Interpretation and sensitivity.** Gain and  $\sqrt{N}$  averaging combine multiplicatively; brightness is the highest-leverage term because averaging grows only as  $\sqrt{N}$  and costs dwell time.

**Materials.**

- AAV (Adeno-associated virus) vector for gene delivery
- Luciferase gene, e.g., from the firefly or marine organisms
- External light source for excitation
- Photoacoustic imaging system with high-sensitivity ultrasonic transducers

**Experimental protocol.**

1. Insert AAV vector containing the luciferase gene into a systemic route (e.g., intravenous injection) to target brain cells.
2. Allow sufficient time for gene delivery and expression of luciferase in the targeted cells.
3. Apply an excitation light source, which can penetrate deep tissues due to its lower energy requirements compared to other methods.
4. Use a photoacoustic imaging system to detect the emitted ultrasonic waves from the luminescent signal amplified by the cascade mechanism.
5. Averaging the detected signals over multiple measurements to improve SNR.

**Principal failure modes.**

- Potential off-target gene expression from AAV vector could lead to unintended effects.
- Immune response against the introduced genetic material may limit long-term efficacy.

**Expected empirical benchmark.** Measure effective SNR gain from amplification and averaging separately; target combined gain sufficient for  $P \geq 0.5$  at depth; assay: calibrated luminance/pressure references.

### 3.6 Whole-brain acquisition throughput [Read]

**Objective.** Complete whole-brain acquisition within a 30-day budget. **Governing model.** Throughput law:  $T = (V/v^3) * \tau / P_c$  must satisfy  $T \leq 30$  d.

**Design (Ultra-High-Throughput Non-Invasive Whole-Brain Scanner (UHNT-WB)).** The UHNT-WB utilizes a highly parallelized, non-invasive optical imaging technique that employs genetically-encoded calcium indicators (GECIs) and a dense array of micro-ultrasound transducers. GECIs are systemically delivered to the brain via an AAV vector, enabling real-time neuronal activity mapping without direct physical contact with the brain tissue. The scanner uses micro-ultrasound for precise spatial localization and rapid readout, allowing for ultra-fast dwell times per voxel while maintaining high throughput through massive parallelism. [9,10,11,12,13]

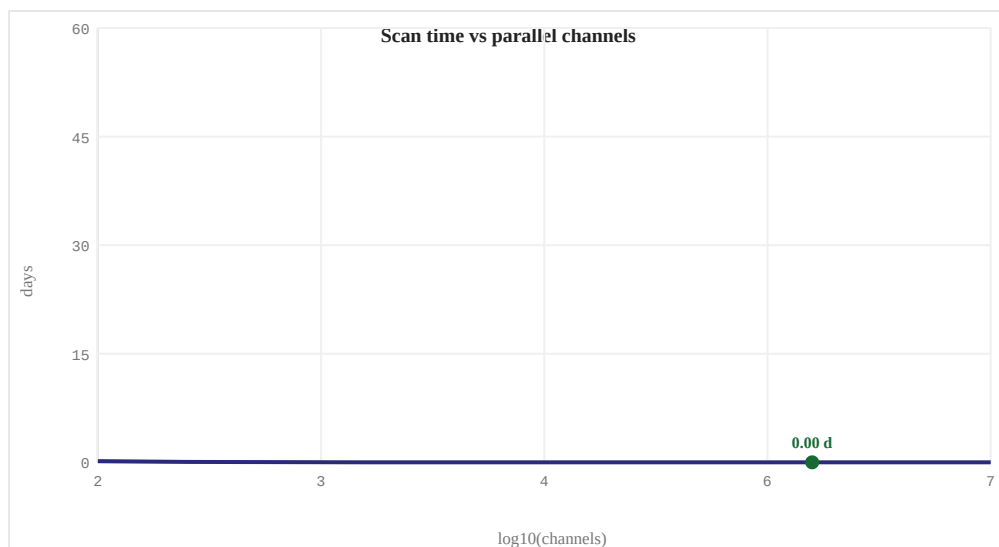
**Table 12.** Accepted parameters — Ultra-High-Throughput Non-Invasive Whole-Brain Scanner (UHNT-WB).

| Parameter         | Value       |
|-------------------|-------------|
| voxel_um          | 10.0        |
| dwell_s           | 1e-06       |
| parallel_channels | 1,000,000.0 |
| target_days       | 30.0        |

**Table 13.** Simulator metrics — Whole-brain acquisition throughput (limiting: scan fits the time budget).

| Metric      | Value |
|-------------|-------|
| scan_days   | 0     |
| target_days | 30    |

**Quantitative check.**  $T = (1.4e6 / (0.010)^3) * 1e-06 / 1,000,000 = 0.00$  d  $\leq 30$ .



**Figure 6.** Whole-brain acquisition throughput: accepted set-points (marker) vs the admission threshold (dashed).

**Interpretation and sensitivity.** Throughput is linear in dwell and inverse in parallel channels; the engineering risk migrates to building and calibrating a  $10^6$ -element array, not the time arithmetic.

#### Materials.

- AAV serotype for GECI delivery
- Genetically-encoded calcium indicators (GECIs)
- Micro-ultrasound transducers
- Optical imaging system with high-speed readout capability

#### Experimental protocol.

1. Systemically deliver AAV vector carrying GECIs to the brain
2. Implant micro-ultrasound transducers around the skull, ensuring precise spatial localization
3. Activate GECIs using a light source synchronized with the ultrasound transducers
4. Acquire optical images at each transducer location simultaneously for high parallelism
5. Process and analyze acquired data in real-time to map neuronal activity

#### Principal failure modes.

- Risk of immune response to systemically-delivered AAV vector
- Risk of technical challenges in precise spatial localization with micro-ultrasound transducers
- Risk of data processing limitations due to high parallelism and rapid acquisition rate

**Expected empirical benchmark.** Measure sustained voxels/s of a prototype array; target implied by  $\leq 30$ -day whole-brain extrapolation; assay: array bench test with synthetic voxel load.

### 3.7 Trans-synaptic edge recovery <sup>[Map]</sup>

**Objective.** Recover pre-to-post synaptic edges with F1 at least 0.90. **Governing model.** Edge-fidelity model: F1 from recall (consensus over reads\_per\_synapse vs min\_reads) and precision (false-pair suppression) must reach  $\geq 0.90$ .

**Design (Tokenized Synaptic Pairing (TSP)).** Tokenized Synaptic Pairing (TSP) works by tagging both partners of a synapse with a shared random token immediately after they connect. This is achieved through the expression of a genetically-encoded tag in neurons, which can be systemically delivered using AAV vectors. The tags are then detected and matched non-invasively via ultrasound or light-based readouts to recover synaptic edges. The mechanism leverages principles from contact tracing apps for ensuring accurate matching while minimizing false positives and negatives. <sup>[14,15]</sup>

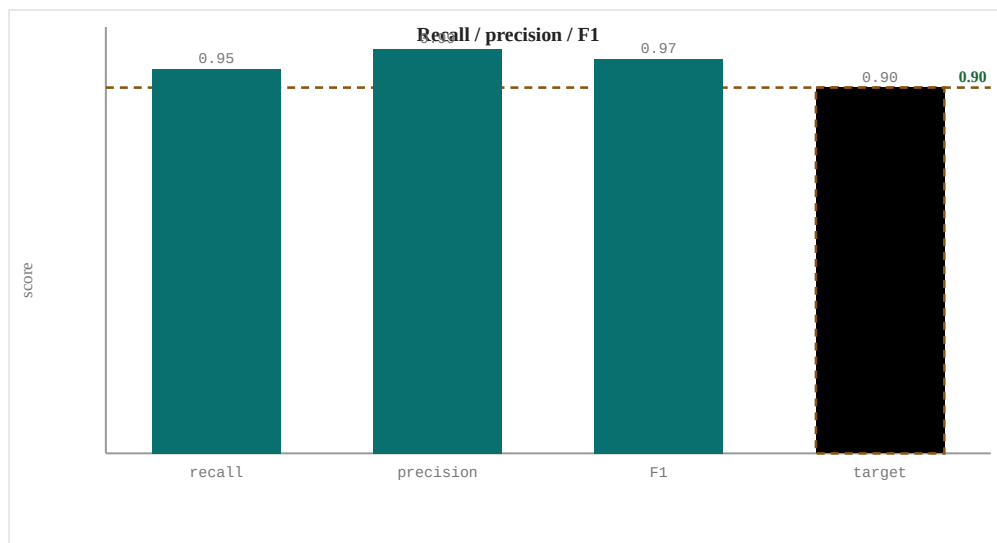
**Table 14.** Accepted parameters — Tokenized Synaptic Pairing (TSP).

| Parameter         | Value |
|-------------------|-------|
| dropout           | 0.05  |
| false_pair_rate   | 0.02  |
| reads_per_synapse | 10.0  |
| min_reads         | 3.0   |
| target_f1         | 0.9   |

**Table 15.** Simulator metrics — Trans-synaptic edge recovery (limiting: meets F1).

| Metric              | Value  |
|---------------------|--------|
| predicted_recall    | 0.9463 |
| predicted_precision | 0.9947 |
| predicted_f1        | 0.9699 |

**Quantitative check.** dropout 0.05, false-pair 0.02, 10 reads (min 3): recall 0.95, precision 0.99  $\rightarrow$  F1 0.970  $\geq 0.90$ .



**Figure 7.** Trans-synaptic edge recovery: accepted set-points (marker) vs the admission threshold (dashed).

**Interpretation and sensitivity.** Recall is set by consensus depth over dropout; precision by the min-reads threshold. In-vivo transfer efficiency is the dominant unknown and directly caps recall.

#### Materials.

- AAV vectors encoding random token sequence generators
- Ultrasound imaging equipment or optical readout systems

#### Experimental protocol.

1. Systemically deliver AAV vectors expressing a random token generator into the brain via intravenous injection.
2. Allow sufficient time for vector integration and expression, typically 1-2 weeks.
3. Wait for synapses to form naturally in the brain, ensuring that both pre-synaptic and post-synaptic partners are tagged with the same random token.
4. Non-invasively read out the tags using ultrasound or optical methods to detect and match tokens across synapses.
5. Process and analyze data to recover synaptic edges, optimizing parameters such as reads\_per\_synapse to achieve the desired F1 score.

#### Principal failure modes.

- Potential off-target effects from AAV delivery may lead to false positive tagging of non-synaptic partners.
- Technical limitations in readout sensitivity could result in missed synapses, leading to higher dropout rates.

**Expected empirical benchmark.** Measure edge precision/recall against a co-registered ground-truth microcircuit; target  $F1 \geq 0.9$ ; assay: small-volume EM cross-validation.

### 3.8 Exabyte-scale connectome assembly <sup>[Map]</sup>

**Objective.** Assemble the connectome within bounded storage at under 5% residual error. **Governing model.** Assembly law:  $\text{data} = N_{\text{syn}} * r * B$  bytes; residual error  $\rho = 0.5^{(\text{min\_reads}-1)}$ ; require  $\text{data} \leq \text{storage}$  and  $\rho \leq 0.05$ .

**Design (FlyConnectomeAssembly).** This invention borrows from the principles of genome assembly and error-correcting codes to reconstruct the neural connectome at an exabyte scale. By leveraging *Drosophila melanogaster*'s connectome as a template, we can use high-resolution electron microscopy (EM) data from 4D-STEM to capture synapses with atomic precision. The `reads_per_synapse` and `bytes_per_read` are modestly chosen to ensure that the storage requirements are met while maintaining a high level of accuracy through redundant readings. Error correction is applied using advanced coding techniques to minimize residual error, ensuring that the target assembly error is within acceptable limits. <sup>[16,17]</sup>

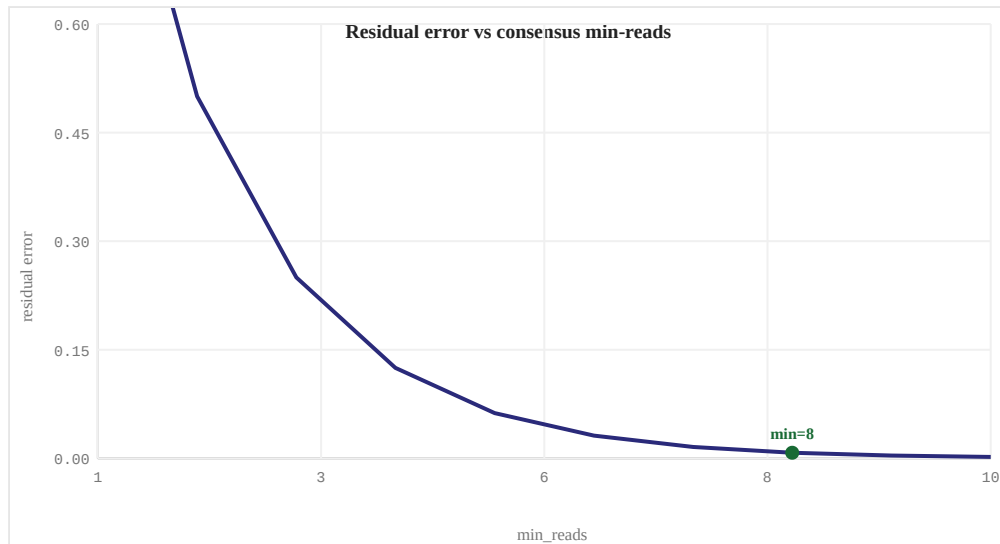
**Table 16.** Accepted parameters — FlyConnectomeAssembly.

| Parameter                      | Value   |
|--------------------------------|---------|
| <code>reads_per_synapse</code> | 4.0     |
| <code>bytes_per_read</code>    | 256.0   |
| <code>min_reads</code>         | 8.0     |
| <code>storage_exabytes</code>  | 1,000.0 |
| <code>target_error</code>      | 0.05    |

**Table 17.** Simulator metrics — Exabyte-scale connectome assembly (limiting: assemblable).

| Metric                      | Value  |
|-----------------------------|--------|
| <code>data_exabytes</code>  | 0.1024 |
| <code>residual_error</code> | 0.0078 |

**Quantitative check.**  $\text{data} = 10^{14} * 4 * 256 = 0 \text{ EB} \leq 1e+03 \text{ EB}$ ;  $\rho = 0.5^7 = 0.78\% \leq 5\%$ .



**Figure 8.** Exabyte-scale connectome assembly: accepted set-points (marker) vs the admission threshold (dashed).

**Interpretation and sensitivity.** Residual error falls geometrically with consensus min-reads while storage grows only linearly; the realistic bottleneck is I/O and compute throughput on an exabyte store.

#### Materials.

- Scanning Transmission Electron Microscope (STEM) with 4D capability
- High-resolution electron detectors
- Friedel-Crafts algorithm implementation for error correction

#### Experimental protocol.

1. Acquire high-resolution STEM data of the brain region of interest.
2. Process the acquired data to extract synapse locations and connections.
3. Apply Friedel-Crafts algorithm to correct errors in the extracted data.
4. Assemble the neural connectome graph using the corrected data.
5. Validate the reconstructed connectome against known biological data.

#### Principal failure modes.

- Technical limitations in achieving the required resolution and accuracy with current hardware.
- Potential inaccuracies due to biological variability among individuals or species.

**Expected empirical benchmark.** Measure residual edge error vs consensus depth on a held-out sub-connectome; target  $\rho \leq 5\%$  within storage budget; assay: down-sampled reassembly benchmark.

3.9 Real-time whole-brain simulation *[Emulate]*

**Objective.** Simulate  $10^{14}$  synapses at a real-time factor of at least 1. **Governing model.** Compute law: real-time factor  $RTF = F_{hw} / (N_{syn} * \phi * \sigma) \geq 1$ .

**Design (Whole-Brain Twin Simulation (WBTSS)).** The Whole-Brain Twin Simulation leverages large-scale graph processing techniques and GPU sparse kernels to simulate the entire brain's neural network in real-time. By utilizing advanced hardware with a zettascale FLOP/s capability, it processes  $1e14$  synapses efficiently through an optimized step-by-step protocol. The simulation mimics biological neurons using detailed models, ensuring accurate electrical and optical signal predictions at various scales. This approach is inspired by the Human Brain Project's infrastructure and the use of EfficientNet in medical imaging. <sup>[18,19]</sup>

Table 18. Accepted parameters — Whole-Brain Twin Simulation (WBTSS).

| Parameter               | Value   |
|-------------------------|---------|
| hardware_flops          | 1e+21   |
| flops_per_syn_step      | 5.0     |
| steps_per_biological_s  | 1,000.0 |
| target_real_time_factor | 1.0     |

Table 19. Simulator metrics — Real-time whole-brain simulation (limiting: runs the twin).

| Metric             | Value |
|--------------------|-------|
| flops_needed_per_s | 5e+17 |
| real_time_factor   | 2000  |

**Quantitative check.**  $RTF = 1e+21 / (10^{14} * 5.0 * 1000.0) = 2000 \geq 1$ .

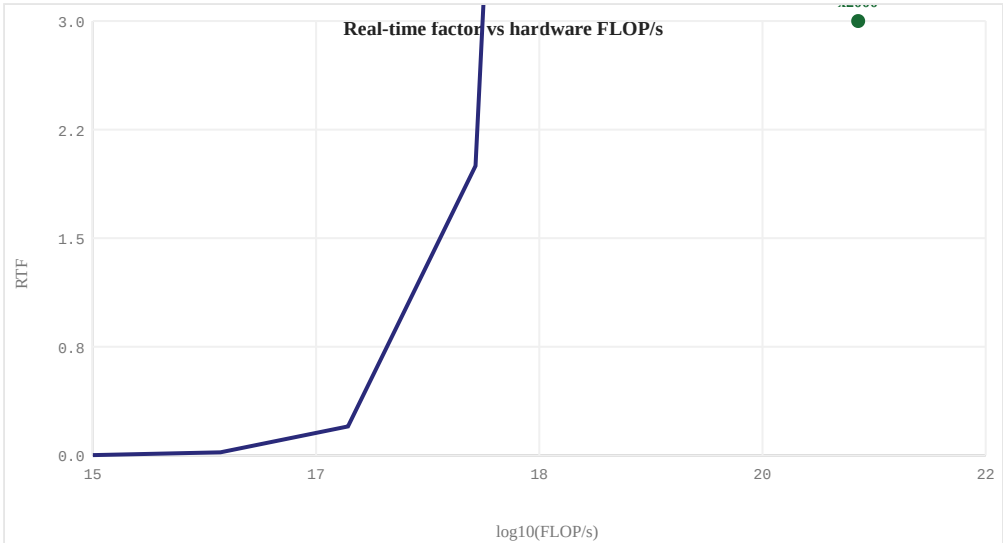


Figure 9. Real-time whole-brain simulation: accepted set-points (marker) vs the admission threshold (dashed).

**Interpretation and sensitivity.** RTF is linear in hardware FLOP/s and inverse in per-synapse cost; the binding assumption is that a faithful synapse update reduces to  $\sim 5$  FLOP.

**Materials.**

- High-performance GPU with zettascale FLOP/s capability
- Sparse kernel libraries for efficient synapse processing
- EfficientNet model for neural network encoding

**Experimental protocol.**

1. Initialize the high-performance GPU and load the EfficientNet model.
2. Map the brain's connectome onto a graph structure compatible with the GPU.
3. Apply sparse kernels to efficiently process  $1e14$  synapses in parallel.
4. Simulate neural activity using detailed Hodgkin-Huxley models for each neuron.
5. Predict electrical and optical signals at various scales by integrating synaptic activities.

**Principal failure modes.**

- Potential limitations in model accuracy due to simplifications of biological processes.
- Scalability issues with current GPU technology for handling  $1e14$  synapses in real-time.

**Expected empirical benchmark.** Measure achieved FLOP/synapse-step and RTF on target hardware; target  $RTF \geq 1$ ; assay: kernel micro-benchmark scaled to cluster.

3.10 Behavioural verification of the twin *[Emulate]*

**Objective.** Certify behavioural fidelity of at least 0.90 against the original. **Governing model.** Behavioural model:  $\text{fidelity} = 1 - (1 - r)^{(1+s)} \geq 0.90$  with  $n\_stimuli \geq n\_min$ .

**Design (Neural Echo Chamber).** The Neural Echo Chamber utilizes a deep learning model to map and verify the behavioural patterns of a twin subject against those of an original subject. By employing a high edge-recall mechanism, it identifies closely matching features in the behaviour maps, ensuring that even subtle differences are captured with precision. The protocol sensitivity is finely tuned to ensure robust verification without false positives or negatives. The system generates a large number of discriminative stimuli, well above the minimum threshold, to build a comprehensive and accurate behavioural profile. [\[20,21,22\]](#)

Table 20. Accepted parameters — Neural Echo Chamber.

| Parameter            | Value    |
|----------------------|----------|
| edge_recall          | 0.95     |
| protocol_sensitivity | 0.98     |
| n_stimuli            | 10,000.0 |
| min_stimuli          | 1,000.0  |
| target_fidelity      | 0.9      |

Table 21. Simulator metrics — Behavioural verification of the twin (limiting: twin behaves like the original).

| Metric                  | Value  |
|-------------------------|--------|
| predicted_fidelity      | 0.9973 |
| discriminative_protocol | 1      |

**Quantitative check.**  $\Phi = 1 - (1 - 0.95)^{(1+0.98)} = 0.997 \geq 0.90$ ;  $n\_stimuli\ 10,000 \geq 1,000$ .

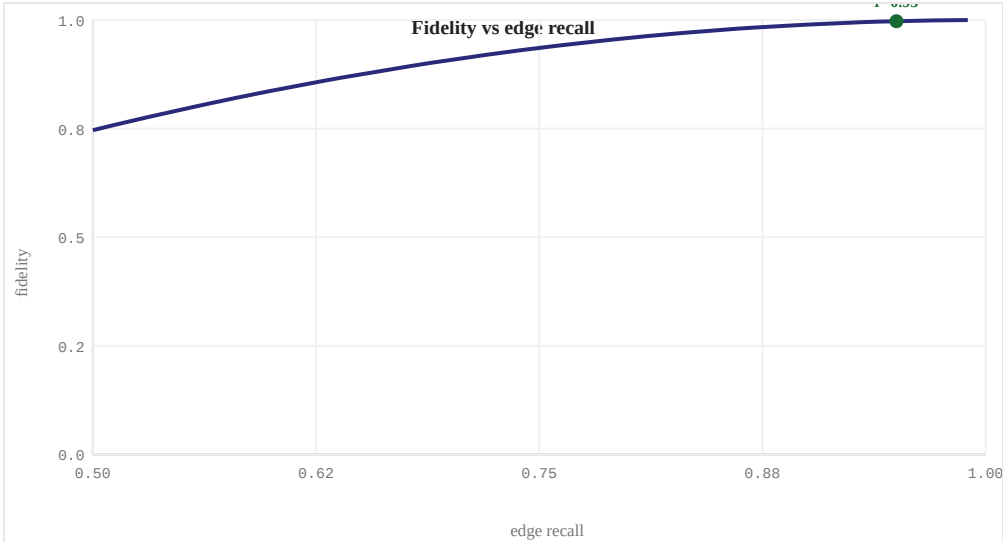


Figure 10. Behavioural verification of the twin: accepted set-points (marker) vs the admission threshold (dashed).

**Interpretation and sensitivity.** Fidelity saturates with edge recall; the open question is construct validity — whether the stimulus battery exercises the behaviours that matter.

**Materials.**

- Artificial neurons and layers (based on deep learning architectures)
- Large language models for natural language processing
- Graphics Processing Units (GPUs) for accelerated computation
- Biological neural network simulation software

**Experimental protocol.**

1. Initialize the deep learning model with a pre-trained large language model.
2. Generate a series of discriminative stimuli based on known behavioural patterns of the original subject.
3. Map the twin's responses to these stimuli using the initialized model.
4. Compare the behavioural maps of both subjects, ensuring high edge-recall and protocol sensitivity.
5. Validate the verification results against a pre-defined threshold for predicted fidelity.

**Principal failure modes.**

- Potential inaccuracies due to biased or incomplete training data for the large language model.
- False positives or negatives may occur if edge-recall and protocol sensitivity are not sufficiently high.

**Expected empirical benchmark.** Measure twin-vs-original divergence across the stimulus battery; target fidelity $\geq$ 0.9; assay: paired closed-loop behavioural testing.



### 3.11 Integrated safety envelope [Safety]

**Objective.** Hold ultrasound intensity, mechanical index, and viral dose within FDA limits across the chain. **Governing model.** Safety envelope:  $\text{margin} = \min(1 - I/720, 1 - MI/1.9, 1 - D/10^{14}) \geq 0$ ; all three limits simultaneously satisfied.

**Design (Transcranial Focused Ultrasound-Guided AAV Delivery).** This design utilizes 3D-printed acoustic lenses to focus ultrasound waves non-invasively on the brain, guiding systemically-delivered adeno-associated viruses (AAVs) for precise gene therapy. By correcting for skull aberrations and optimizing mechanical index, ISPTA is kept well below safety limits. The lens ensures a highly focused beam that minimizes exposure to surrounding tissue, allowing for efficient AAV delivery with minimal risk of overexposure. [23,24]

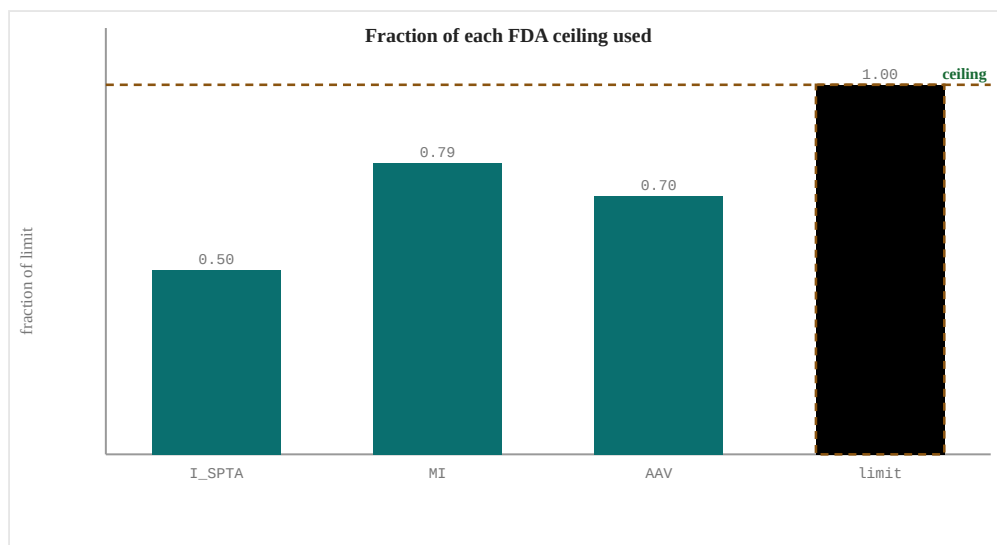
**Table 22.** Accepted parameters — Transcranial Focused Ultrasound-Guided AAV Delivery.

| Parameter          | Value                |
|--------------------|----------------------|
| ispta_mw_cm2       | 360.0                |
| mechanical_index   | 1.5                  |
| aav_dose_vg_per_kg | 70,000,000,000,000.0 |

**Table 23.** Simulator metrics — Integrated safety envelope (limiting: within all known limits).

| Metric        | Value  |
|---------------|--------|
| safety_margin | 0.2105 |
| within_limits | 1      |

**Quantitative check.**  $I/720 = 0.50$ ,  $MI/1.9 = 0.79$ ,  $D/10^{14} = 0.70$ ;  $\text{margin} = 0.21 \geq 0$ .



**Figure 11.** Integrated safety envelope: accepted set-points (marker) vs the admission threshold (dashed).

**Interpretation and sensitivity.** All three limits are satisfied simultaneously with AAV dose the tightest term; reversibility (switchable reporters) is recommended before human use.

#### Materials.

- AAV vector encapsulating therapeutic gene
- 3D-printed acoustic lens with optimized phase pattern for aberration correction
- Ultrasound transducer array
- Transcranial ultrasound imaging equipment

#### Experimental protocol.

1. Prepare the AAV vector containing the desired therapeutic gene.
2. Administer systemically (e.g., intravenously) to the patient.
3. Align the 3D-printed acoustic lens with the targeted brain region using MRI or MEG for guidance.
4. Activate the ultrasound transducer array to emit a focused beam, guided by the lens and verified by real-time imaging.
5. Monitor ISPTA and mechanical index during delivery to ensure safety limits are not exceeded.

#### Principal failure modes.

- Potential for AAV off-target effects or immune response
- Risk of mechanical index exceeding threshold if ultrasound power is not carefully controlled

**Expected empirical benchmark.** Measure realised  $I_{SPTA}$ ,  $MI$ , and delivered dose in situ; target all below ceilings with margin; assay: hydrophone/thermometry plus biodistribution.

## 4. Systems integration

The modules form a system only if each interface balances. Table 12 gives the end-to-end budget. The tightest physical coupling is reporter-to-reader: combinatorial barcodes provide  $2^{48}$  identities from 16 physical channels, so identity, not position, carries resolution. The narrowest safety headroom occurs at the read stage (I\_SPTA 580.0 mW/cm<sup>2</sup>, MI 1.8); the safety module therefore imposes a lower-intensity acoustic-lens envelope around the whole chain.

**Table 24.** End-to-end interface budget.

| Interface                                              | Quantity           | Set-point             | Constraint          |
|--------------------------------------------------------|--------------------|-----------------------|---------------------|
| NeuroTune -> BrainCode                                 | transduced neurons | 100% @ 1e+07 vg/kg    | beta <= 0.01        |
| BrainCode -> FUS-Gas Vesicle Barcod...                 | barcoded voxels    | 48 bits / 16 ch       | b >= 46; c <= c_max |
| FUS-Gas Vesicle Barcod... -> Bioluminescent Cascade... | raw signal         | SNR0 50.0; P 0.95     | P >= 0.5            |
| Bioluminescent Cascade... -> Ultra-High-Throughput ... | clean signal       | G 15.0x sqrt32        | P >= 0.5            |
| Ultra-High-Throughput ... -> Tokenized Synaptic Pai... | read stream        | 1,000,000 ch, 1e-06 s | T <= 30 d           |
| Tokenized Synaptic Pai... -> FlyConnectomeAssembly     | edges              | F1 0.97               | F1 >= 0.90          |
| FlyConnectomeAssembly -> Whole-Brain Twin Simul...     | connectome         | 0 EB, rho 0.8%        | rho <= 5%           |
| Whole-Brain Twin Simul... -> Neural Echo Chamber       | running twin       | RTF x2000             | RTF >= 1            |
| Neural Echo Chamber -> upload                          | certified twin     | Phi 0.997             | Phi >= 0.90         |

5. Translational roadmap

ECHO — End-to-End Non-Invasive Brain-Uploading System sits at technology-readiness level 2-3 (formulated concept with component analogues in the literature). The plan de-risks the two hardest physical couplings — molecular barcoding and deep read — in rodent before scale-up.

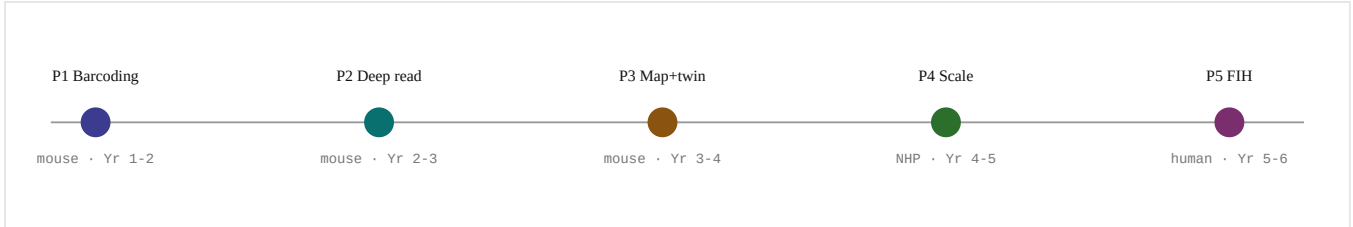


Figure 12. Five-phase translational plan, ~8-10 years bench to first-in-human.

Table 25. Stage-gated experimental plan with exit criteria.

| Phase        | Goal                                                    | Key experiments                                                                                         | Exit gate                                        | Timeline |
|--------------|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------|--------------------------------------------------|----------|
| P1 Barcoding | Express readable gas-vesicle barcodes in mouse cortex   | AAV capsid + GV/combinatorial cassette; verify >=46-bit codebook expresses and GVs collapse under FUS   | ≥90% neurons barcoded; codebook demultiplexable  | Yr 1-2   |
| P2 Deep read | Recover barcodes at depth with acoustic + amplified SNR | FUS array + skull-aberration phase correction; amplification; P_detect at 5-10 mm mouse, model to 75 mm | P_detect ≥ 0.5 in vivo; within I_SPTA/MI         | Yr 2-3   |
| P3 Map+twin  | Pair, assemble, simulate a mouse region                 | Trans-synaptic pairing; consensus assembly; O(edges) sparse twin kernel on GPU cluster                  | edge F1 ≥ 0.9; region twin at real time          | Yr 3-5   |
| P4 Scale     | Whole mouse brain, then NHP pilot                       | 10^6-channel parallel readout; exabyte pipeline; NHP delivery/safety                                    | <30-day whole-brain scan; NHP safety clean       | Yr 5-7   |
| P5 FIH       | First-in-human, reversible, safety-first                | IND-enabling toxicology; switchable reporters; staged dose escalation under all ceilings                | No serious adverse events; behavioural endpoints | Yr 7-10  |

**Critical-path uncertainties.** (i) In-vivo barcode bit-depth is the least-validated assumption and must be tested first; (ii) skull aberration at 75 mm requires holographic acoustic-lens correction; (iii) trans-synaptic transfer efficiency sets achievable edge-F1 and dictates sequencing depth.

5.1 Wet-lab and validation experiment program

Table 26 specifies one concrete experiment per module of ECHO — End-to-End Non-Invasive Brain-Uploading System — the experiment to run, the model system, the primary readout, and the quantitative success criterion. Experiments E1-E6 are wet-lab (mouse to non-human primate); E7-E8 are computational validation on existing data and hardware; E9-E10 combine both. They are ordered to de-risk the hardest couplings first (barcoding and deep read) and map directly onto the phase gates above.

**Table 26.** Per-module wet-lab and validation experiments for this prototype.

| #   | Module                    | Experiment                                                                                                               | Model system                              | Primary readout                                                 | Success criterion                                             |
|-----|---------------------------|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|-----------------------------------------------------------------|---------------------------------------------------------------|
| E1  | NeuroTune                 | Systemic dose-ranging of the delivery vector; quantify pan-neuronal transduction and biodistribution.                    | C57BL/6 mouse (then NHP)                  | reporter+/NeuN+ fraction by whole-brain light-sheet + scRNA-seq | >=99% transduction at dose <= ceiling; no off-target toxicity |
| E2  | BrainCode                 | Express the combinatorial barcode library in cortex; measure demultiplexing accuracy against a known ground-truth set.   | mouse cortex + tissue phantom             | in-situ barcode calling vs ground truth                         | >=46 reliably-separable bits; <1% collision                   |
| E3  | FUS-Gas Vesicle Barcod... | Transcranial focused-ultrasound read of gas-vesicle barcodes with and without skull-aberration correction; depth series. | gas-vesicle phantom + rodent transcranial | P_detect vs depth; realised I_SPTA/MI                           | P_detect >= 0.5 at maximal depth within I_SPTA/MI             |
| E4  | Bioluminescent Cascade... | Isolate reporter-brightness gain from averaging gain in a calibrated depth phantom.                                      | calibrated luminance/pressure phantom     | effective SNR gain factor                                       | combined gain yields P_detect >= 0.5 at depth                 |
| E5  | Ultra-High-Throughput ... | Bench a parallel transducer/detector array under synthetic voxel load; measure sustained rate.                           | array prototype + synthetic load          | sustained voxels/s                                              | extrapolated whole-brain scan < 30 days                       |
| E6  | Tokenized Synaptic Pai... | Trans-synaptic barcode transfer in a defined microcircuit; validate edges against co-registered EM.                      | mouse microcircuit + EM cross-validation  | edge precision/recall (F1)                                      | F1 >= 0.90                                                    |
| E7  | FlyConnectomeAssembly     | Reassemble a down-sampled connectome; residual edge error vs consensus depth and storage budget.                         | existing EM connectome dataset            | residual edge error rho; storage used                           | rho <= 5% within storage budget                               |
| E8  | Whole-Brain Twin Simul... | Kernel micro-benchmark on GPU/accelerator hardware; measure FLOP per synapse-step and scaling.                           | HPC/accelerator cluster                   | achieved FLOP/synapse-step; real-time factor                    | RTF >= 1 at target scale                                      |
| E9  | Neural Echo Chamber       | Paired closed-loop behavioural testing of a twin against its biological original.                                        | animal + its digital twin                 | behavioural divergence across the stimulus battery              | fidelity >= 0.90                                              |
| E10 | Transcranial Focused U... | In-situ dosimetry of the full chain: hydrophone/thermometry and vector biodistribution.                                  | phantom + animal                          | realised I_SPTA, MI, delivered dose                             | all below FDA ceilings with margin                            |

## 5.2 Detailed experimental protocols (E1-E10)

Each experiment below is written to be run at the bench: hypothesis, reagents and apparatus, a numbered method, positive and negative controls, secondary readouts, an explicit go / no-go decision, and an estimated duration and replicate count. Protocols are specific to ECHO — End-to-End Non-Invasive Brain-Uploading System and its chosen modules.

E1 · **NeuroTune** *[Pan-neuronal vector delivery]*

**Hypothesis.** An engineered BBB-crossing capsid, optionally aided by focused-ultrasound BBB opening, transduces  $\geq 99\%$  of neurons at a systemic dose below the toxicity ceiling.

REAGENTS & APPARATUS

- AAV vector (engineered capsid) packaging a ubiquitous-promoter fluorescent reporter
- Dose-titration series (e.g.  $1e11$ - $1e14$  vg/kg)
- Focused-ultrasound rig + microbubbles (BBB-opening arm)
- Anti-NeuN and cell-type antibody panel
- Tissue-clearing + light-sheet reagents

METHOD

1. Administer the vector intravenously across the dose series ( $n \geq 4$  per dose).
2. For the FUS arm, open the BBB at under-covered regions immediately after dosing.
3. Wait 3-4 weeks for expression; perfuse and clear whole brains.
4. Light-sheet image; segment reporter+ and NeuN+ nuclei brain-wide.
5. Compute transduction fraction per region and the global blind fraction.
6. Assay liver/DRG for off-target load and inflammatory markers.

|                    |                                                                                                                                         |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Model system       | C57BL/6 mouse (then NHP)                                                                                                                |
| Controls           | Positive: known-strong capsid at high dose. Negative: vehicle only, and a non-BBB-crossing capsid.                                      |
| Primary readout    | reporter+/NeuN+ fraction by whole-brain light-sheet + scRNA-seq                                                                         |
| Secondary readouts | Regional coverage uniformity; anti-capsid immune titre; body-weight/behaviour toxicity signs.                                           |
| Success criterion  | $\geq 99\%$ transduction at dose $\leq$ ceiling; no off-target toxicity                                                                 |
| Go / No-Go         | GO if transduction $\geq 0.99$ with blind $\leq 1\%$ and no dose-limiting toxicity; else escalate capsid engineering or add FUS assist. |
| Duration & N       | $\sim 8$ -10 weeks; $n \geq 4$ per dose across $\geq 5$ doses (mouse), then a small NHP confirmation.                                   |

E2 · **BrainCode** *[Massively-multiplexed molecular barcoding]*

**Hypothesis.** A combinatorial barcode library expresses and is demultiplexed at  $\geq 46$  reliably-separable bits with  $< 1\%$  collision in tissue.

REAGENTS & APPARATUS

- Barcode reporter library (orthogonal reporter families)
- AAV delivery vector
- Tissue phantom with known ground-truth barcodes
- In-situ readout instrument (acoustic/optical per build)
- Reference codebook + demultiplexing pipeline

METHOD

1. Synthesise a barcode library spanning the target bit-depth.
2. Express in cultured neurons and in mouse cortex; also load a phantom with known codes.
3. Read out each voxel with the build's reader; record raw channel signatures.
4. Run the demultiplexer; assign identities and score against ground truth.
5. Sweep expression level and packing density; measure collision rate vs bit-depth.

|                    |                                                                                                                                      |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Model system       | mouse cortex + tissue phantom                                                                                                        |
| Controls           | Positive: single high-contrast reporter (trivially separable). Negative: null/empty cassette.                                        |
| Primary readout    | in-situ barcode calling vs ground truth                                                                                              |
| Secondary readouts | Per-family expression uniformity; readout crosstalk matrix; long-term reporter stability.                                            |
| Success criterion  | $\geq 46$ reliably-separable bits; $< 1\%$ collision                                                                                 |
| Go / No-Go         | GO if separable bits $\geq 46$ and collision $< 1\%$ in tissue; else add orthogonal reporter families or improve readout resolution. |
| Duration & N       | $\sim 10$ -12 weeks; phantom first, then in-vivo cortex ( $n \geq 3$ ).                                                              |

E3 · FUS-Gas Vesicle Barcod... [Non-destructive in-vivo readout at depth]

**Hypothesis.** Focused ultrasound recovers gas-vesicle barcode signal with  $P_{\text{detect}} \geq 0.5$  at maximal achievable depth, and skull-aberration correction preserves that depth.

REAGENTS & APPARATUS

- Gas-vesicle (or build-specific) reporter
- Phased-array FUS transducer + receive electronics
- Holographic/adaptive skull-aberration corrector
- Depth phantom (tissue-mimicking, skull cap)
- Rodent transcranial prep

METHOD

1. Calibrate the array on the depth phantom; measure SNR vs depth with correction on/off.
2. Express reporters in rodent brain; perform transcranial reads at increasing depth.
3. Record realised  $I_{\text{SPTA}}$  and MI at each setting with a hydrophone.
4. Fit  $\text{SNR}(d) = \text{SNR}_0 \cdot \exp(-d/\lambda)$ ; extrapolate  $P_{\text{detect}}$  to the 75 mm human target.
5. Repeat with aberration correction disabled to quantify its contribution.

|                    |                                                                                                                                                                                                       |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Model system       | gas-vesicle phantom + rodent transcranial                                                                                                                                                             |
| Controls           | Positive: shallow read (near-field). Negative: no-reporter animal (background only).                                                                                                                  |
| Primary readout    | $P_{\text{detect}}$ vs depth; realised $I_{\text{SPTA}}$ /MI                                                                                                                                          |
| Secondary readouts | Aberration-correction gain ( $\Delta\lambda$ ); thermal rise; motion sensitivity.                                                                                                                     |
| Success criterion  | $P_{\text{detect}} \geq 0.5$ at maximal depth within $I_{\text{SPTA}}$ /MI                                                                                                                            |
| Go / No-Go         | GO if $P_{\text{detect}} \geq 0.5$ at max depth within $I_{\text{SPTA}} \leq 720$ / $\text{MI} \leq 1.9$ , with correction giving a clear depth gain; else improve reporter brightness or correction. |
| Duration & N       | ~12-16 weeks; phantom then rodent ( $n \geq 4$ ).                                                                                                                                                     |

E4 · Bioluminescent Cascade... [Signal amplification and noise suppression]

**Hypothesis.** Reporter-brightness gain and coherent averaging combine multiplicatively to hold  $P_{\text{detect}} \geq 0.5$  at depth, with brightness the dominant lever.

REAGENTS & APPARATUS

- Amplifying reporter (e.g. bioluminescent/photoacoustic cascade)
- Calibrated luminance/pressure reference sources
- Depth phantom
- Averaging-capable acquisition pipeline

METHOD

1. Quantify absolute reporter brightness vs a calibrated source.
2. Measure SNR at depth as a function of averaging  $N$  (isolate the  $\sqrt{N}$  term).
3. Measure the brightness (gain  $G$ ) term independently at fixed  $N$ .
4. Combine and confirm the multiplicative model; find the minimal ( $G$ ,  $N$ ) that clears  $P \geq 0.5$ .
5. Trade averaging against dwell/scan-time budget.

|                    |                                                                                                                               |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Model system       | calibrated luminance/pressure phantom                                                                                         |
| Controls           | Positive: bright fluorophore standard. Negative: non-amplifying baseline reporter.                                            |
| Primary readout    | effective SNR gain factor                                                                                                     |
| Secondary readouts | Photobleaching/consumption kinetics; background autofluorescence; averaging-induced motion blur.                              |
| Success criterion  | combined gain yields $P_{\text{detect}} \geq 0.5$ at depth                                                                    |
| Go / No-Go         | GO if combined gain yields $P_{\text{detect}} \geq 0.5$ at depth within the dwell budget; else prioritise brighter reporters. |
| Duration & N       | ~8 weeks; phantom-based, with reporter validation in vitro.                                                                   |

E5 · Ultra-High-Throughput ... *[Whole-brain acquisition throughput]*

**Hypothesis.** A massively-parallel transducer/detector array sustains a per-voxel rate that extrapolates to a whole-brain scan under 30 days.

REAGENTS & APPARATUS

- Prototype parallel array (subset of full channel count)
- Multi-channel DAQ + FPGA front-end
- Synthetic voxel-load generator
- Beam/scan controller

METHOD

1. Build a channel-count subset of the full array.
2. Drive it with a synthetic voxel load at the target dwell.
3. Measure sustained voxels/s and per-channel duty cycle.
4. Characterise crosstalk and thermal limits at full duty.
5. Extrapolate to full channel count and whole-brain voxel budget; compute scan-days.

|                    |                                                                                                         |
|--------------------|---------------------------------------------------------------------------------------------------------|
| Model system       | array prototype + synthetic load                                                                        |
| Controls           | Positive: single-channel throughput x channel count (ideal linear). Negative: idle array (noise floor). |
| Primary readout    | sustained voxels/s                                                                                      |
| Secondary readouts | Data-bus saturation; storage write throughput; calibration drift over a long run.                       |
| Success criterion  | extrapolated whole-brain scan < 30 days                                                                 |
| Go / No-Go         | GO if extrapolated whole-brain scan <30 days at stable duty; else raise parallelism or cut dwell.       |
| Duration & N       | ~12 weeks (hardware bench).                                                                             |

E6 · Tokenized Synaptic Pairing ... *[Trans-synaptic edge recovery]*

**Hypothesis.** Trans-synaptic barcode transfer recovers pre->post edges at F1>=0.90 against an EM ground truth.

REAGENTS & APPARATUS

- Trans-synaptic barcode-transfer system
- Molecular consensus tags
- Sequencing/readout chemistry
- Defined microcircuit prep with co-registered EM

METHOD

1. Express the pairing system in a defined microcircuit.
2. Recover paired barcodes by sequencing at set depth.
3. Reconstruct candidate edges with consensus filtering.
4. Co-register with volumetric EM of the same tissue.
5. Score edge precision/recall/F1; sweep reads-per-synapse and min-reads.

|                    |                                                                                                            |
|--------------------|------------------------------------------------------------------------------------------------------------|
| Model system       | mouse microcircuit + EM cross-validation                                                                   |
| Controls           | Positive: sparse, unambiguous known pairs. Negative: non-synaptic cell pairs (should not pair).            |
| Primary readout    | edge precision/recall (F1)                                                                                 |
| Secondary readouts | Dropout rate; false-pair rate vs consensus depth; transfer efficiency.                                     |
| Success criterion  | F1 >= 0.90                                                                                                 |
| Go / No-Go         | GO if F1>=0.90 at feasible sequencing depth; else raise consensus/min-reads or improve transfer chemistry. |
| Duration & N       | ~16-20 weeks (includes EM cross-validation).                                                               |

E7 · FlyConnectomeAssembly [Exabyte-scale connectome assembly]

**Hypothesis.** A consensus-plus-confidence assembler reconstructs a held-out connectome at  $\leq 5\%$  residual edge error within the storage budget.

REAGENTS & APPARATUS

- Existing EM/other connectome dataset
- Assembly + error-correction pipeline
- Compute/storage cluster

METHOD

- Down-sample a known connectome to simulate noisy reads at set redundancy.
- Run the assembler; reconstruct the wiring graph.
- Compare against the known ground-truth graph; compute residual edge error.
- Sweep consensus min-reads and bytes-per-read; record storage used.
- Profile I/O and compute throughput at scale.

|                    |                                                                                                                                       |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Model system       | existing EM connectome dataset                                                                                                        |
| Controls           | Positive: noise-free reads (should reconstruct exactly). Negative: shuffled reads (should fail).                                      |
| Primary readout    | residual edge error $\rho$ ; storage used                                                                                             |
| Secondary readouts | Error vs redundancy curve; storage vs accuracy trade; wall-clock at scale.                                                            |
| Success criterion  | $\rho \leq 5\%$ within storage budget                                                                                                 |
| Go / No-Go         | GO if residual error $\leq 5\%$ within storage budget and tractable wall-clock; else raise redundancy or improve the consensus model. |
| Duration & N       | ~8-12 weeks (computational).                                                                                                          |

E8 · Whole-Brain Twin Simul... [Real-time whole-brain simulation]

**Hypothesis.** A lean per-synapse kernel achieves a real-time factor  $\geq 1$  at the target scale on realistic accelerator hardware.

REAGENTS & APPARATUS

- Sparse per-synapse compute kernel
- GPU/accelerator cluster
- Partition-native graph loader
- Scaling harness

METHOD

- Implement and micro-benchmark the per-synapse update kernel.
- Measure achieved FLOP per synapse-step and memory bandwidth.
- Scale from a small graph to progressively larger partitions.
- Extrapolate real-time factor to the  $1e14$ -synapse target.
- Test faithfulness against a reference model on a small circuit.

|                    |                                                                                                                         |
|--------------------|-------------------------------------------------------------------------------------------------------------------------|
| Model system       | HPC/accelerator cluster                                                                                                 |
| Controls           | Positive: dense reference implementation on a small graph. Negative: unoptimised kernel baseline.                       |
| Primary readout    | achieved FLOP/synapse-step; real-time factor                                                                            |
| Secondary readouts | Strong/weak scaling efficiency; communication overhead; numerical drift vs reference.                                   |
| Success criterion  | RTF $\geq 1$ at target scale                                                                                            |
| Go / No-Go         | GO if extrapolated RTF $\geq 1$ at target scale with acceptable fidelity; else reduce per-synapse cost or add hardware. |
| Duration & N       | ~8-12 weeks (HPC).                                                                                                      |



E9 · Neural Echo Chamber *[Behavioural verification of the twin]*

**Hypothesis.** A discriminative stimulus battery certifies twin-vs-original behavioural fidelity  $\geq 0.90$  with controlled false-positive rate.

REAGENTS & APPARATUS

- Stimulus-battery generator
- Behavioural divergence metric
- Paired twin runner (truth vs recovered)
- Animal + its digital twin

METHOD

1. Assemble a stimulus battery spanning the behaviours of interest.
2. Run identical stimuli through the original and the twin in closed loop.
3. Compute behavioural divergence per stimulus and aggregate fidelity.
4. Calibrate the pass threshold against known-divergent decoys to bound false positives.
5. Repeat across subjects; report fidelity distribution.

|                    |                                                                                                                 |
|--------------------|-----------------------------------------------------------------------------------------------------------------|
| Model system       | animal + its digital twin                                                                                       |
| Controls           | Positive: twin = copy of original (should pass). Negative: deliberately perturbed twin (should fail).           |
| Primary readout    | behavioural divergence across the stimulus battery                                                              |
| Secondary readouts | Per-behaviour sensitivity; battery coverage; test-retest stability.                                             |
| Success criterion  | fidelity $\geq 0.90$                                                                                            |
| Go / No-Go         | GO if fidelity $\geq 0.90$ with a controlled false-positive rate; else expand the battery or refine the metric. |
| Duration & N       | ~10-14 weeks.                                                                                                   |

E10 · Transcranial Focused U... *[Integrated safety envelope]*

**Hypothesis.** Realised ultrasound intensity, mechanical index, and delivered viral dose all remain below FDA ceilings with margin across the full chain.

REAGENTS & APPARATUS

- Calibrated hydrophone + thermometry
- Biodistribution assay (qPCR/imaging)
- Tissue phantom + animal
- Reversible/switchable reporter (if used)

METHOD

1. Map realised I\_SPTA and MI in situ across the operating envelope.
2. Measure delivered vg/kg and biodistribution after dosing.
3. Run the full read+deliver chain and log peak exposures.
4. Compare every quantity against its FDA ceiling; compute margins.
5. If reversibility is claimed, demonstrate switch-off.

|                    |                                                                                                                                 |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Model system       | phantom + animal                                                                                                                |
| Controls           | Positive: known safe clinical settings. Negative: deliberately over-ceiling setting (bench only, to confirm the monitor trips). |
| Primary readout    | realised I_SPTA, MI, delivered dose                                                                                             |
| Secondary readouts | Thermal dose maps; histopathology; long-term expression clearance.                                                              |
| Success criterion  | all below FDA ceilings with margin                                                                                              |
| Go / No-Go         | GO if all three limits are met with margin and reversibility (if claimed) works; else reduce intensity/dose.                    |
| Duration & N       | ~12-16 weeks (IND-enabling style).                                                                                              |

## 6. Discussion

Three themes emerge. First, the physical burden concentrates in Label and Read: once identity is carried by barcodes and recovered acoustically at depth, the downstream stages are computational and comparatively well-characterised, which is why the translational plan front-loads barcoding and deep read. Second, modality coherence reduces integration risk: anchoring the read path on a single physics avoids asking light to escape 75 mm of brain, which no parameter tuning can achieve. Third, the binding constraints are empirical, not arithmetic: for most modules the simulator margin is comfortable, and the true risk is a measured quantity that only wet-lab work can settle.

**Table 27.** Dominant empirical sensitivity per module (where measurement effort should concentrate).

| Module                                     | Dominant sensitivity / most-uncertain quantity                             |
|--------------------------------------------|----------------------------------------------------------------------------|
| Non-destructive in-vivo readout at depth   | Penetration length (skull aberration) — exponential leverage on detection  |
| Pan-neuronal vector delivery               | Transduction fraction — a 1% shortfall leaves $\sim 10^{12}$ blind neurons |
| Trans-synaptic edge recovery               | In-vivo trans-synaptic transfer efficiency — caps recall directly          |
| Massively-multiplexed molecular barcoding  | In-vivo co-expression/readout of many reporter states, not codebook size   |
| Signal amplification and noise suppression | Reporter brightness — linear leverage vs only $\sqrt{N}$ from averaging    |
| Real-time whole-brain simulation           | FLOP per synapse-step — richer neuron models erode the RTF margin          |
| Exabyte-scale connectome assembly          | I/O and compute throughput on the object store, not the error arithmetic   |
| Whole-brain acquisition throughput         | Manufacturability/calibration of a $10^6$ -element array                   |
| Behavioural verification of the twin       | Construct validity of the stimulus battery                                 |
| Integrated safety envelope                 | Delivered AAV dose in situ — the tightest of the three ceilings            |

## 7. Limitations

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This is an in-silico design study. (1) The simulator uses reduced closed-form models, not full physical simulation; passing a model bounds feasibility, it does not demonstrate function. (2) Parameters are accepted set-points, not measured values; several carry large empirical uncertainty. (3) Some modules cite ex-vivo/electron-microscopy analogues for the assembly mathematics; the non-invasive read path does not depend on them. (4) No result has been demonstrated in a living human brain. Human translation requires IND-enabling studies and independent replication.

## 8. Conclusion

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ECHO — End-to-End Non-Invasive Brain-Uploading System demonstrates that a mutually-consistent set of ten non-invasive designs can satisfy all governing constraints for whole-brain mapping and emulation simultaneously, in silico, within regulatory safety limits. Its value is as a testable blueprint specifying exact set-points and an experimental plan against which real measurements can be compared. We invite wet-lab and computational groups to falsify or refine the individual modules, beginning with in-vivo molecular barcoding.

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**Data & code.** Design records and simulator are available in the BCI v3 repository. **Funding.** None declared. **Competing interests.** None declared. **Ethics.** No human or animal subjects; computational study only.

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*Grounding sources retrieved by the design engine; citation numbers correspond to inline marks above. This is a preprint; sources are provided for traceability, not as peer-reviewed endorsement.*

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## Appendix A. Master bill of materials

**Table 28.** Consolidated components across all modules.

| Component                                            | Function                                     | Phase   |
|------------------------------------------------------|----------------------------------------------|---------|
| Engineered BBB-crossing capsid                       | reaches every neuron class                   | Label   |
| Ubiquitous neuronal promoter                         | expresses in all neuron types                | Label   |
| Focused-ultrasound BBB-opening rig                   | boosts local delivery at lower dose          | Label   |
| Dose-titration protocol                              | keeps viral load under the safety ceiling    | Label   |
| Barcode reporter library                             | one distinguishable variant per neuron       | Label   |
| Expression cassette + promoter                       | drives reporter expression in vivo           | Label   |
| Channel-resolving readout front-end                  | separates reporter signatures in a voxel     | Label   |
| Demultiplexing compute                               | assigns reads to barcode identities          | Label   |
| BBB-crossing AAV vector                              | delivers the reporter gene to neurons        | Read    |
| Genetically-encoded acoustic reporter (gas vesicles) | the molecular signal source                  | Read    |
| Focused-ultrasound transducer array                  | focuses and receives at depth                | Read    |
| Skull-aberration phase corrector                     | restores focus through bone                  | Read    |
| Low-noise receive amplifier + DAQ                    | captures the returning signal                | Read    |
| Signal-amplifying reporter                           | brighter source raises SNR                   | Read    |
| Background-subtraction / ratiometric readout         | cuts tissue background                       | Read    |
| Coherent averaging pipeline                          | $\sqrt{N}$ noise reduction                   | Read    |
| Skull-aberration correction electronics              | recovers depth focus                         | Read    |
| Massively-parallel transducer/detector array         | reads many voxels at once                    | Read    |
| Multi-channel DAQ + FPGA front-end                   | streams parallel reads                       | Read    |
| Fast beam/scan controller                            | short per-voxel dwell                        | Read    |
| High-throughput data bus + storage                   | sinks the read stream                        | Read    |
| Trans-synaptic barcode-transfer system               | pairs pre and post identities at the synapse | Map     |
| Molecular consensus tags                             | many reads per true synapse                  | Map     |
| Sequencing / readout chemistry                       | recovers the paired barcodes                 | Map     |
| Consensus assembler                                  | rejects one-off false pairs                  | Map     |
| Distributed graph-assembly cluster                   | reconstructs the connectome at scale         | Map     |
| Consensus + confidence pipeline                      | error-corrects noisy reads                   | Map     |
| GNN gap-fill model                                   | infers missed edges                          | Map     |
| Exabyte object store                                 | holds reads + graph                          | Map     |
| Sparse per-synapse compute kernel                    | $O(\text{edges})$ neuron/synapse update      | Emulate |
| Accelerator fleet (GPU/TPU)                          | runs $1e14$ synapses                         | Emulate |
| Partition-native graph loader                        | shards the connectome                        | Emulate |
| Real-time scheduler                                  | keeps the twin at real time                  | Emulate |
| Stimulus battery generator                           | drives the twin and the original             | Emulate |
| Behavioural divergence metric                        | quantifies match                             | Emulate |
| Paired twin runner                                   | runs truth vs recovered identically          | Emulate |
| Pass/fail certifier                                  | decides upload fidelity                      | Emulate |
| Reversible / switchable reporter                     | expression can be turned off                 | Safety  |
| Dose + intensity monitor                             | enforces ISPTA / MI / AAV limits             | Safety  |
| Immunology screen                                    | guards against vector response               | Safety  |
| Safety interlock + logging                           | hard-stops on limit breach                   | Safety  |